

Dept. of Computer Science and Engineering, KUET

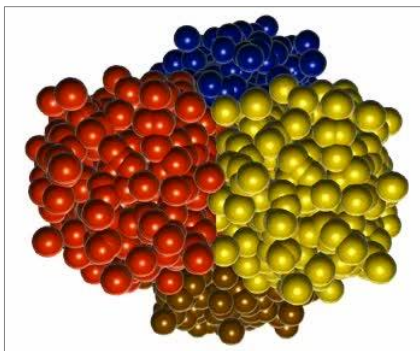
Postgraduate Program, January Semester, 2026

Programming Assignment on Advanced Digital Image Processing (CSE 6243)

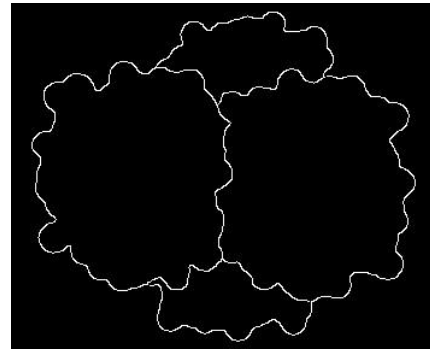
Assignment # 02

Topic: Image Segmentation

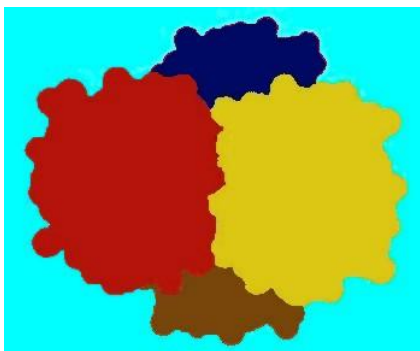
1. Manually prepare Ground Truth (GT) images by yourself for the supplied image dataset. You may use LabelMe or any other annotation software.
 - a. Provide the GT images as contour as well as in mask format. Also provide a contour / mask overlay image. See example below.
 - b. Be sure to save the images in png or bmp format. JPEG format may introduce compression artifacts.



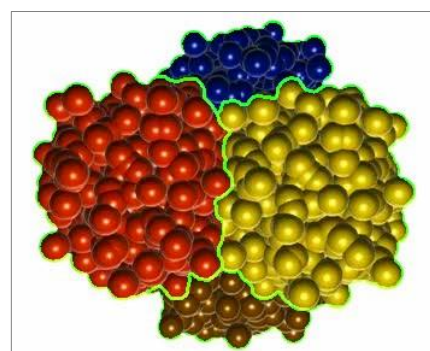
Input



Contour



Mask



Contour Overlay

2. Propose an idea of converting the image into some feature space. You should use each of the features (color space (RGB and Lab/HSV), spatial position, and textures) separately and their various combinations.
 - a. For Texture extraction you may use Gabor filter bank/Leung-Malik filter bank or any other method. Visualize the filters and their convolution result on the images.
 - b. You may directly use the filter bank result as feature space or you may create some texton dictionary.

- c. Please study some related works. Here will be the novelty of your idea. Objective of the method will be better segmentation.
3. Write a program (in Matlab or C++/Python-OpenCv) to segment each of the image represented by above feature spaces. Use following algorithms
 - a. **K-means clustering** (use appropriate values of K for each image, e.g. for above image K=5; you may also use other values of K to over-segment)
 - b. **Mean shift Algorithm**
 - i. Use appropriate (at least 2 different value) bandwidth parameter for each feature space
 - ii. Use Gaussian as well as Epanechnikov kernel
4. Apply necessary postprocessing (if required) if there is over segmentation. Compare the machine generated segmentation result with GT image and report the result of different methods
 - a. In a form of normalized confusion matrix and then as Precision, Recall, IoU (intersection over union) and **Boundary F1-score (BF score)** for each image and macro average for all the images. Perform necessary Label Matching before measuring these metrics.
 - b. Use visual / tabular representation of quantitative results
 - c. Also, show all the qualitative results (output images) with proper labelling in the report. When visualizing Gabor filter responses, provide a 'Jet' colormap or normalized heat map to clearly show which textures the filters are activating on
5. Collect GT images from at least one of your peers and test by the best parameter combination of your method and only present the numerical results in the report.
 - a. You should discuss (how you are storing, the annotation tool, etc.) with your peer before creating GT images so that your program smoothly incorporates both of your GT. Don't copy each other GT images and claim that it is yours.
 - b. Blend your and peer's contour images to see the difference.
6. Deliverables
 - a. A detailed report explaining your idea, the visual and quantitative results with proper caption and the program (Don't copy, I'll check it with Turnitin). Use necessary references.
 - b. All the output images (for various parameters), GT images, peers GT and results, annotation files (if any), program files, numerical results of different methods as csv files, and any other things each in separate folder.

- c. Report should be submitted to google classroom with roll no as a separate pdf, the program file as a sperate .m/.c/c++/.py file and all other materials (folders – do not zip) in a single folder.
- d. You are supposed to execute and explain on the next available class after submission.

Note: Program should be written in such a way it will take only two things: an input image and corresponding GT (mask/contour) image or optionally the number of segments required.

- i) Then it will output the segmented image as well as the quantitative results (precision, recall etc.)
- ii) You better create a console menu to show different methods'/parameters results.
- iii) Be sure that outputs will not only be just shown but also be saved in some predefined folder.
- iv) You may resize the image to half for faster output generation.
- v) Choose reasonably size filter bank, Mean shift performs poorly with very high dimensional feature space. You may apply PCA to reduce dimensions while retaining 95% variance before running Mean Shift

Submission date, time: June 11, 2026, mid-day.

Additional info:

- i) Anyone may extend or use this idea on a benchmark dataset and prepare a conference paper.
- ii) Give your best effort to make it a nice project, it will carry maximum weight of the course.
- iii) Feel free to contact me for any clarification.